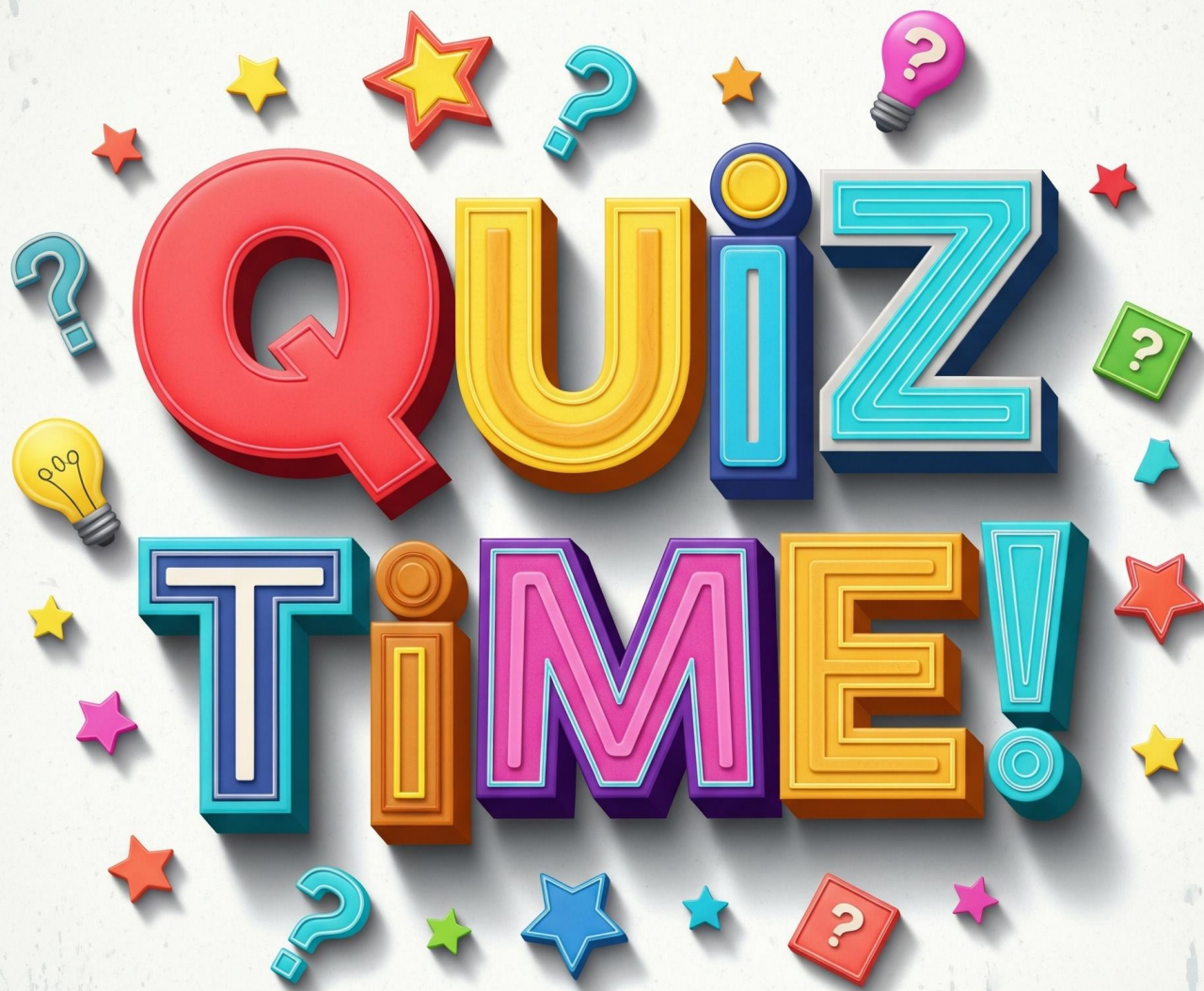
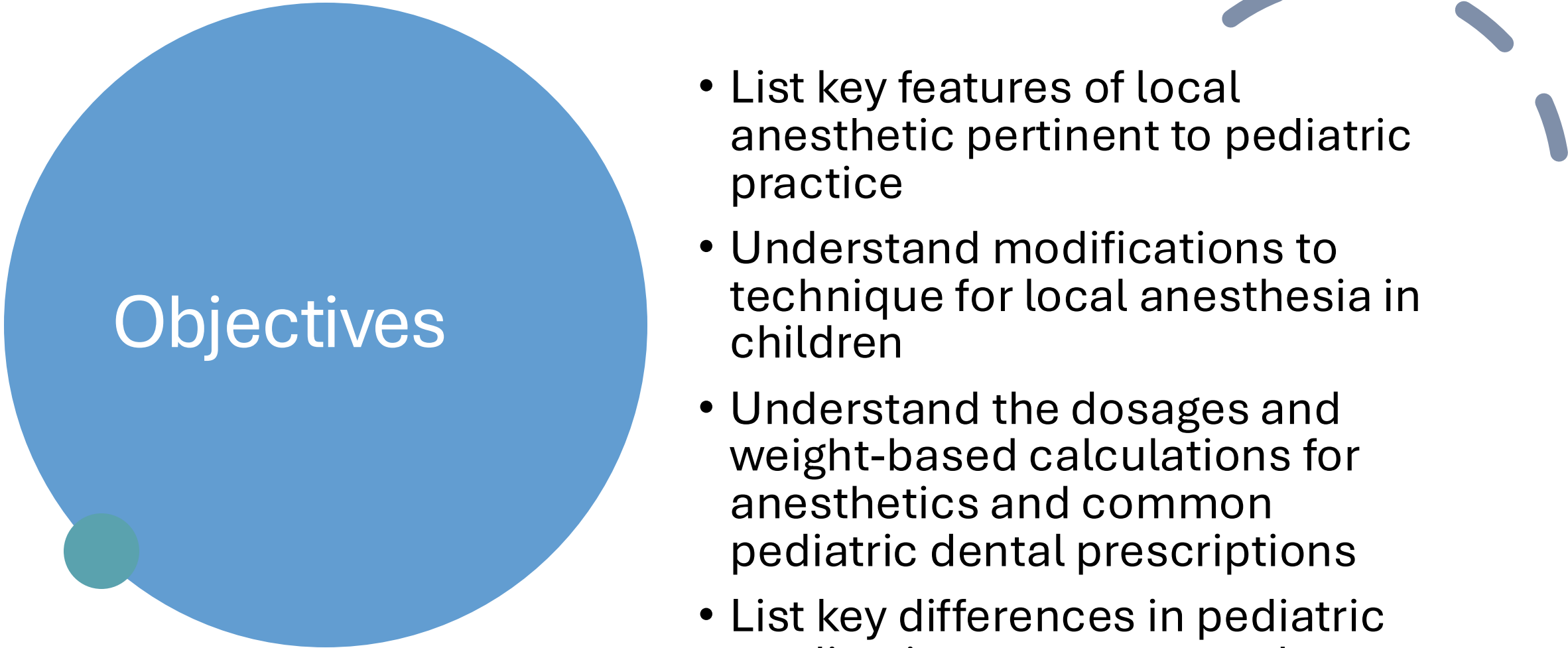






Objectives

- List key features of local anesthetic pertinent to pediatric practice
- Understand modifications to technique for local anesthesia in children
- Understand the dosages and weight-based calculations for anesthetics and common pediatric dental prescriptions
- List key differences in pediatric medication use compared to adult medication use

Outline

Drugs and children

Local anesthetic review

Practice anesthetic calculations in children

Anesthetic complications

Common pediatric medications

Script writing review

Prescription writing activity



Drugs and Children



Is local anesthetic a drug?



Drugs and patients

- Every time you inject - you are prescribing and administering a drug to a patient
- Everything that follows applies to anesthetic, and other prescribed medications

Typical Adult Medications

Lidocaine
500mg max

Amoxicillin 500
mg three times
a day

For adults

We make an assumption that everyone is close enough to the same weight and drug processing ability that we can use the same dose

Children

- Have a huge variety of sizes and weights
- The heaviest 10 year old was 400+ lbs
- The lightest 10 year old was 36 lbs
- We can't give all children the same dose



Children

Dosing in children
is typically based
on weight

Helps us tailor the
dose to the
individual

Even this is not
perfect



Local Anesthetics

Local Anesthetic Function

Unchanged in pediatric
dentistry*

Selective Voltage Gated Sodium
Channel Blocker

Prevents conduction of
sensations along nerve channels

Anesthetic Selection in Pediatrics

Local anesthetic agent and concentration	Band Color				
2 % Lidocaine	Red 185	Red 186	Red 199	Red 200	
3 % Lidocaine	Purple 266	Purple 267			
2 % Mepivacaine	Brown 477	Brown 478	Brown 498	Brown 499	
3 % Mepivacaine	Tan 406	Tan 407	Tan 408		
3 % Prilocaine	Orange 136	Orange 137			
4 % Prilocaine	Yellow 108	Yellow 109	Yellow 110	Yellow 115	Yellow 116
4 % Articaine	Gold 871	Gold 872	Gold 873	Gold 874	Gold 875
0.5 % Bupivacaine	Blue 300	Blue 301			

Local Anesthetic Structure

- Esters
- Amides
- Special





Benzocaine

- Ester type local anesthetic
- Typical Use in Dentistry: Topical Gels
- Metabolized by: Pseudocholinesterase enzymes in the blood



Lidocaine

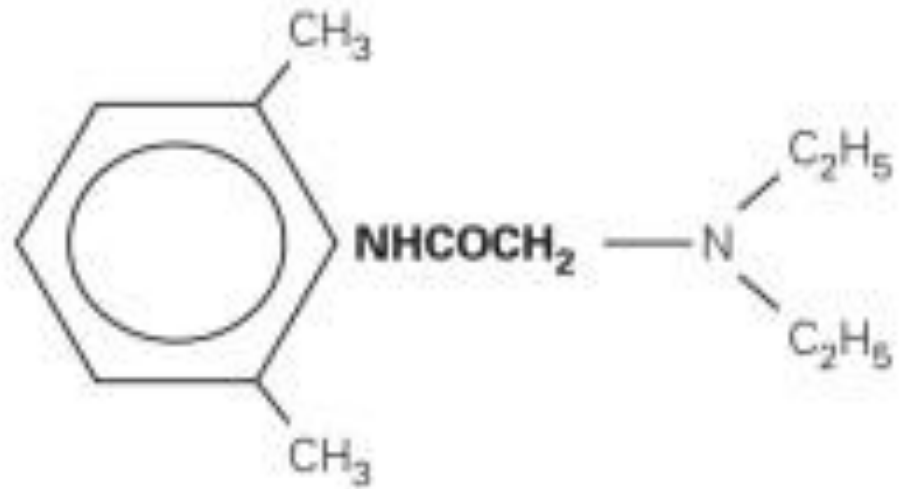
- Amide type local anesthetic
- Typical Use in Dentistry: Injectable Solution
- Metabolized by: Cytochrome p-450 microsomal enzymes in the liver



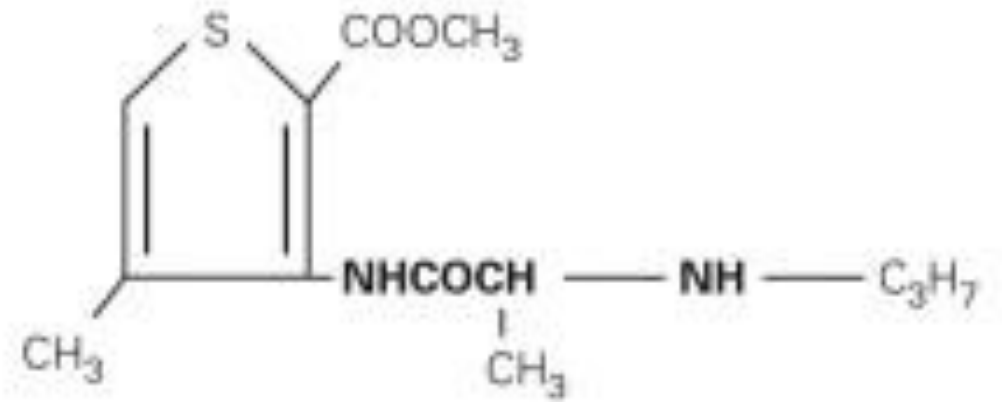
Articaine

- Amide type local anesthetic with an ester side chain
- Typical Use in Dentistry: Injectable Solution
- Metabolized by:
Pseudocholinesterase enzymes in the blood *and*
Cytochrome p-450 microsomal enzymes in the liver





Lidocaine



Articaine



Why does metabolism matter

Metabolism Impacts



DURATION



MAXIMUM DOSE

Anesthetic Selection and Dosing

Local Anesthetic Dosing

Pediatric Dosing is dependent on weight

Each anesthetic has a different maximum dose depending on its structure and metabolism

Lidocaine

- Maximum Dose:
 - 4.4mg/kg
 - 2mg/lbs



Articaine

- Maximum Dose
 - 7mg/kg
 - 3.2 mg/lbs



Others

Mepivacaine

- 4.4 mg/kg
- 2.0 mg/lbs

Prilocaine

- 6.0 mg/kg
- 2.0 mg/kg

Bupivacaine

- 1.3mg/kg
- 0.6mg/lbs

Choosing an Anesthetic

Anesthetic selection depends on

Patient

Procedure

Injection type and location

Patient

Medical history and
allergies

Age

Weight

Procedure

How painful is the procedure expected to be

What depth of anesthesia do we need to achieve

Injection

What technique are we going to use

How much space is there for fluid

What are the local tissue conditions like

2% Lidocaine

- An appropriate anesthetic for all ages
- Appropriate for all injection techniques
- Lower concentration means less anesthetic per carpule



4% Articaine

- Approved for use in age 4 and up
- Appropriate for infiltrations – concerns about paresthesia when used in IANBs
- May penetrate through tissue more effectively
- More efficacious in acidic environments
- Higher concentration means more anesthetic per volume injected



Other agents

- Mepivacaine – rarely used in situations where vasoconstrictor is not desired
 - No vasoconstrictor increases toxicity risk
- Prilocaine – amide type, offers no advantages over lidocaine
- Bupivacaine – Used rarely in children due to long duration of soft tissue anesthesia

Selection

- 2% lidocaine is safe and appropriate in almost all scenarios
- Consider 4% articaine when
 - Infiltration only
 - Expected difficulty achieving anesthetic
 - Smaller volume of injection expected/possible
 - Older than 4 years
- Other anesthetics – rare use cases only

What do you need to know to figure out a max dose

How much a
person can have

- Dose by weight
- Weight

How much we
give

- Concentration
- Volume

Reminders

- Always have a current weight
- Know what you are injecting!
 - Drug
 - Concentration
 - Volume



Hey Dr. Marion.....

- I read that the max dose of lidocaine is 7mg/kg
- Yes, some manufacturers list this now
- The AAP and AAPD currently still recommend the lower max dose of 4.4mg/kg for lidocaine for dental procedures
- This may change in the future
- Lean towards safety
- **For your exam – USE 4.4 MG/KG**



Anesthesia Precautions

Anesthetic Complications

Post-Op

Allergies

Reactions

Overdose

Table 1. INJECTABLE LOCAL ANESTHETICS

Anesthetic	Duration in minutes ^{3,23}				Maximum dosage ²³		Maximum total dosage ^{23*}
	Maxillary infiltration		Mandibular block		mg/kg	mg/lb	
	Pulp	Soft tissue	Pulp	Soft tissue			(mg)
<i>Lidocaine</i>					4.4	2.0	300
2% plain	5		5-10				
2%+1:50,000 epinephrine	60	170	85	190			
2%+1:100,000 epinephrine	60	170	85	190			
<i>Mepivacaine</i>					4.4	2.0	300
3% plain	25	90	40	165			
2%+1:100,000 epinephrine	60	170	85	190			
2%+1:20,000 levonordefrin	50	130	75	185			
<i>Articaine</i>					7.0	3.2	500
4%+1:100,000 epinephrine	60	190	90	230			
<i>Prilocaine</i>					6.0	2.7	400
4% plain	20	105	55	190			
4%+1:200,000 epinephrine	40	140	60	220			
<i>Bupivacaine</i>					1.3	0.6	90
0.5%+1:200,000 epinephrine	40	340	240	440			

Post Operative Biting

Soft tissue
anesthesia lasts
somewhere
between 170-190
minutes

The fastest way a
child tends to
check is by:





Which results in



Allergies

It is possible to be allergic to local anesthetic agents



This used to be more common with ester-type local anesthetics



Most allergic reactions occur to:

Procaine

Tetracaine

Chloroprocaine

Benzocaine

Allergies

More common are reactions to other anesthetic contents - Sodium Bisulfite

It is rare – but documented to have a contact allergy to exogenous epinephrine

Possible that normal action of epi is interpreted as an allergy

Treat all allergies as legitimate until proven otherwise

Allergies in Children

Children are less likely to have encountered an allergen

A dental procedure may be the first time they are exposed to certain agents

Make sure to differentiate between adverse drug actions and true allergies

Reactions

Intravascular
Injections

Methemoglobinemia

Intravascular Injections

Injection into a vessel rapidly
introduces epinephrine into
the circulatory system



Excess Epi can lead to

↑Heart Rate

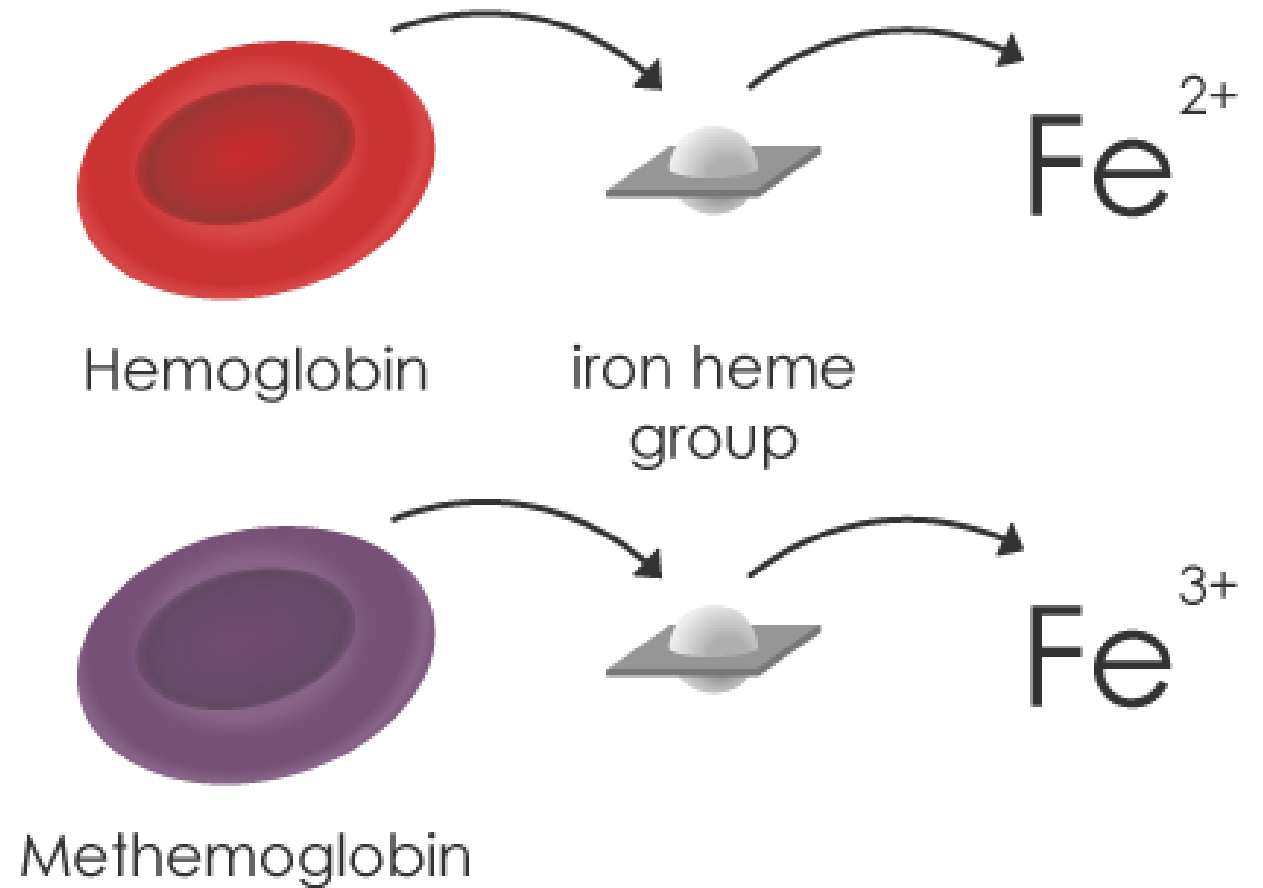
↑Blood
Pressure

Arrhythmias

Uncomfortable
feeling

Methemoglobin

- Hemoglobin in a Ferrous State can bind and release oxygen
- Hemoglobin in a Ferric State binds oxygen tightly (aka Methemoglobin)
- The Ferric version is oxidized, and this occurs spontaneously in the presence of oxidizing compounds



Methemoglobin

- Normally hemoglobin is 97-99% Ferrous and 1-3% Ferric due to the activity of the Methemoglobin reductase enzyme
- If there are an excess of oxidizing compounds in the blood, the enzyme can't keep up
- If methemoglobin reaches 10-20% most people experience symptoms

Methemoglobinemia

- Anesthetic agents and their metabolites can function as strong oxidizing agents – resulting in rapid development of methemoglobin
- Triggering agents
 - Prilocaine
 - Tetracaine
 - Benzocaine
 - Many other drugs

Methemoglobinemia

- Methemoglobinemia results in cyanosis with typical breathing and cardiac function (at the start)
- Cyanosis – blue extremities at the start
- Symptoms develop – Fatigue, Tachypnea, Dyspnea, Tachycardia, Nausea, Vomitting, Dizziness, Confusions, Anxiety
- Symptoms Worsen – Seizure, Coma, Arrhythmia, Hyperlactemia, Death

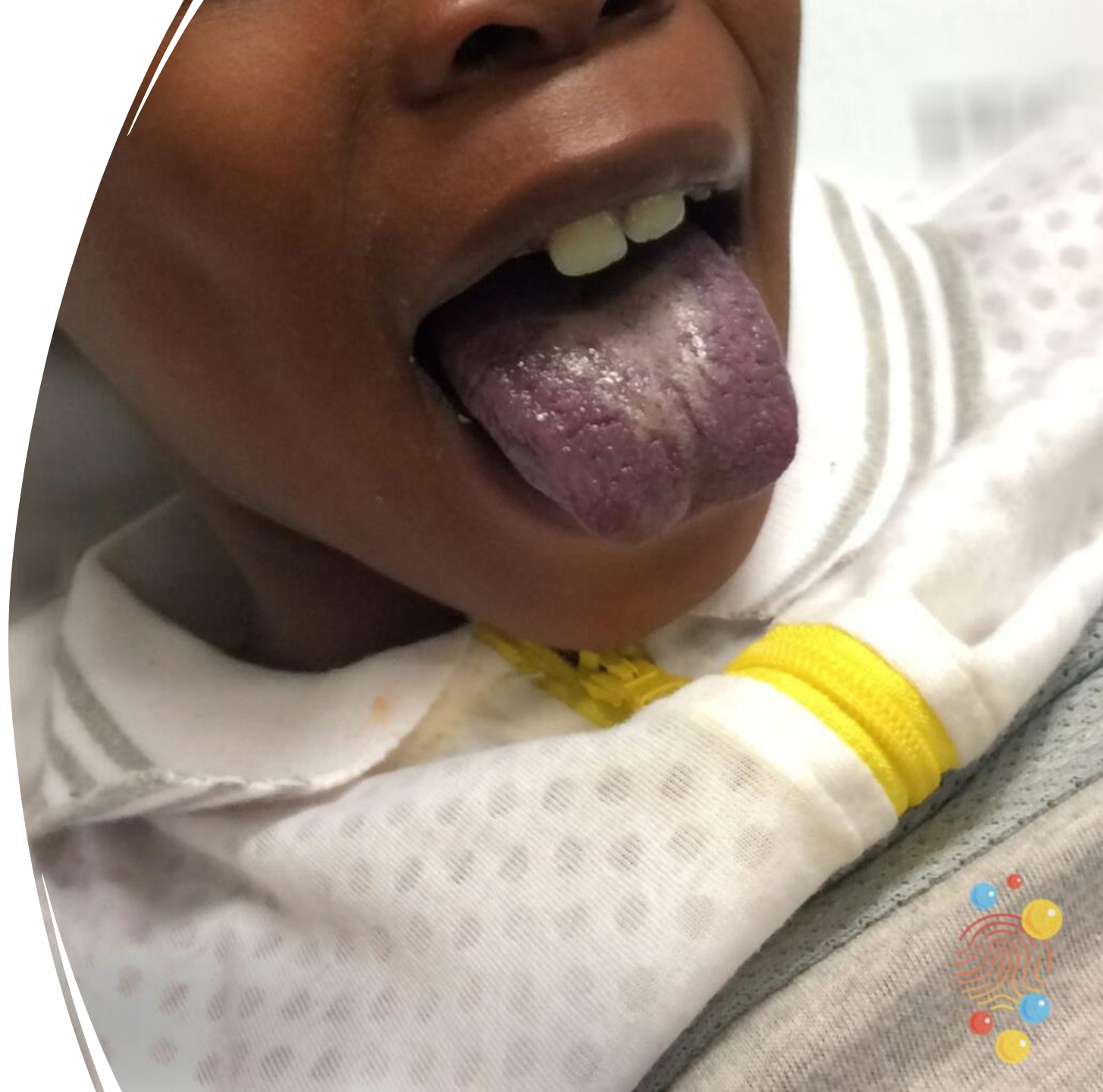
Methemoglobinemia in Children

- Children are more sensitive to changes in oxygenation
- Likely that Methemoglobinemia occurs at lower percentages in younger children and exposure to smaller doses results in larger changes



Methemoglobinemia in Children

- Cyanosis often occurs first in extremities, lips, and tongue
- If the patient has cyanosis with O₂ saturation in the 80s suspect Methemoglobinemia
- Cyanosis is typically observable when O₂ saturation is below 80



Treatment of Methemoglobinemia

- Potentially life threatening condition
- Activate emergency services
- Supplemental oxygen is likely ineffective but administer anyway
- Methylene blue IV

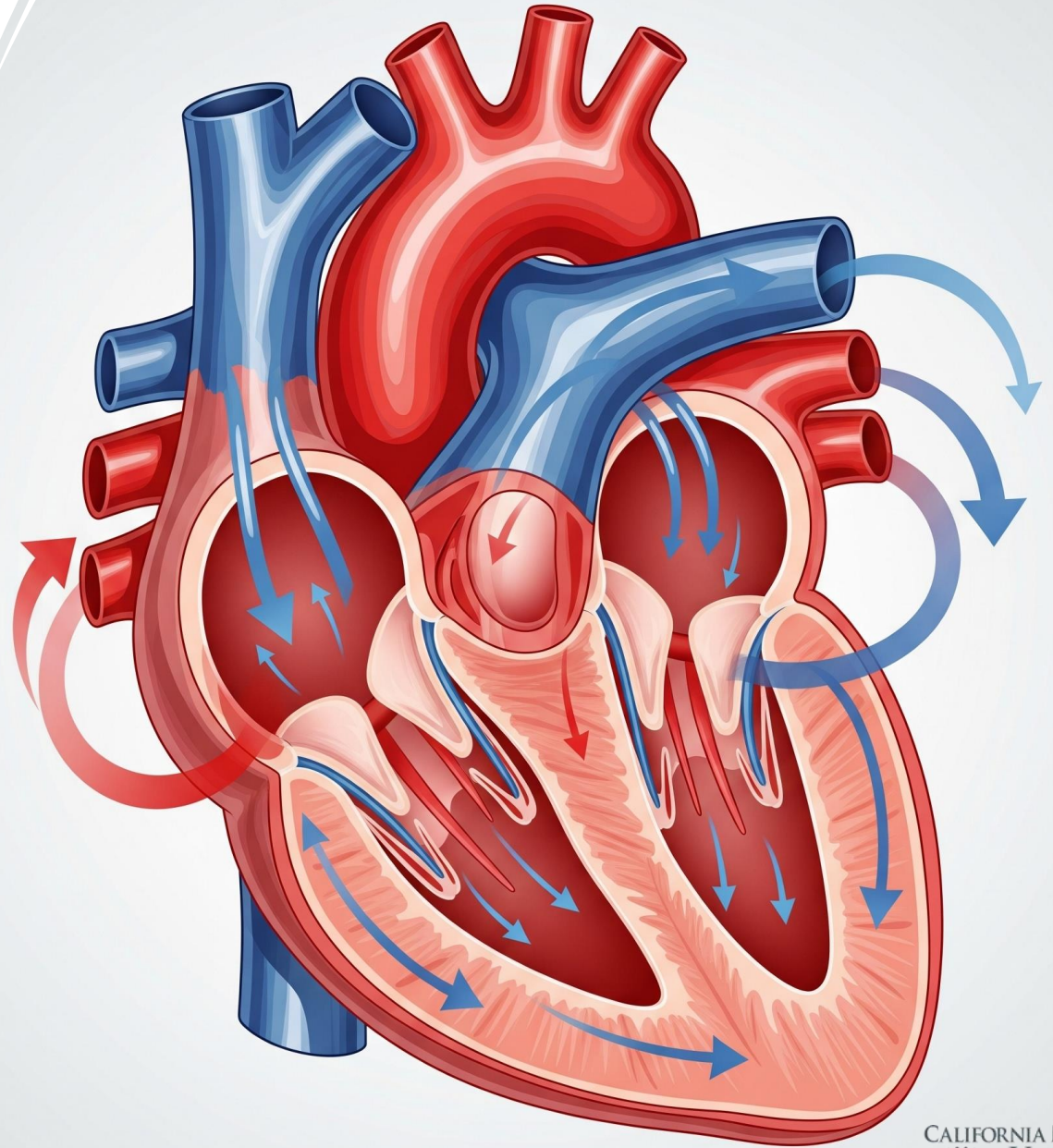


Local Anesthetic Toxicity

- When the amount of drug administered results in undesired negative consequences to the patient
- Local anesthetics tend to result in cardiovascular and CNS effects
- LA has a “biphasic” dose response curve meaning at low dose 1 action occurs, and at higher dose the opposite action occurs

Cardiovascular Toxicity

- Low doses of anesthetic agents have an **antidysrhythmic** action
- Higher doses cause **dysrhythmias**, myocardial depression, and peripheral vasodilation
- Extremely high doses cause severe peripheral dilation and myocardial depression – both of which lead to **Cardiac arrest**



CNS Toxicity

- Low doses cause CNS **excitation** – agitation, talkativeness, irritability
- Further increased dose results in central CNS **depression** – drowsiness, lethargy, and seizures
- Even higher doses cause generalized CNS **depression** – lack of muscle tone, respiratory depression or arrest, and death



Toxicity Management

- The body will continue to metabolize and eliminate local anesthetics
- Keep the patient alive until that can occur
- Active EMS, Supportive Oxygen, and BLS/PALS/ACLS as needed



Toxicity in Children

- The vast majority of local anesthetic toxicity cases occur in children
- Children are not more sensitive to the actions of local anesthetics
- What do you think is the reason?



Toxicity in Children

The #1 cause of toxicity is “too large a dose given”

The lidocaine doses given by dentists treating children ranged from 0.9-**19.3** mg/kg

This is a problem that can be almost entirely avoided

Final Suggestions

Be mindful of needle
safety especially with
children

No procedure is
worth risking toxicity

Medications



Reminder

Dosing by weight is the preferred mechanism for calculating medication for pediatric patients

There are a few exceptions and viable alternative systems

Common Medications for Children

Antibiotics

Analgesics

Antifungals

Analgesics

- Select Agent
- Ibuprofen 40mg/mL suspension
- Children < 12y/o 4-10mg/kg/dose q6h as needed for pain
- Our Patient is still 24 kg
- 10 day course to match with our antibiotics

Calculations

- $24\text{kg} \times (10\text{mg/kg/dose}) = 240\text{ mg/dose}$
- $240\text{mg/dose} (1\text{mL}/40\text{mg}) = 6\text{ mL/dose}$
- $6\text{ mL/dose} * 4\text{ doses/day} = 24\text{ mL/day}$
- $24\text{mL/day} * 10\text{ days} = 240\text{ mL total}$

Script

Rx: Ibuprofen 40mg/mL

Sig: Take 6mL by mouth every 6
hours as needed for pain

Disp: 240mL

Script Writing Suggestions

- Have a good reference and don't be shy about looking things up
- Use a calculator
- Double check if something seems out of place
- Get a weight every time

Dosing for Infants and Children

From Your Healthcare Professional



If possible, use weight to dose; otherwise use age.

mL = milliliter
tsp = teaspoon

DOSE: Every 4 hours as needed. DO NOT GIVE MORE THAN 5 DOSES IN 24 HOURS.



Infants' TYLENOL® Oral Suspension

Active Ingredient:
Acetaminophen 160 mg
(in each 5 mL)

Available in:
 Grape
 Cherry

Use only as directed.



Children's TYLENOL® Oral Suspension

Active Ingredient:
Acetaminophen 160 mg
(in each 5 mL or 1 tsp)

Available in:
 Grape
 Cherry

Use only as directed.

WEIGHT	AGE			
6-11 lbs	0-3 mos	1.25 mL		—
12-17 lbs	4-11 mos	2.5 mL		—
18-23 lbs	12-23 mos	3.75 mL		—
24-35 lbs	2-3 yrs	5 mL		5 mL (1 tsp)
36-47 lbs	4-5 yrs	—	—	7.5 mL (1½ tsp)
48-59 lbs	6-8 yrs	—	—	10 mL (2 tsp)
60-71 lbs	9-10 yrs	—	—	12.5 mL (2½ tsp)
72-95 lbs	11 yrs	—	—	15 mL (3 tsp)

IMPORTANT INSTRUCTIONS for Proper Use

Today's Date: _____

This dosing recommendation from your doctor will expire in 14 DAYS.

- Read and follow the label on all TYLENOL® products.
- Take every 4 hours as needed. Do NOT exceed more than 5 doses in 24 hours.
- Do NOT use with any other product containing acetaminophen.
- Use only the dosing device that comes with a specific product.
- **All Infants' TYLENOL® and Children's TYLENOL® Oral Suspension products have the same acetaminophen concentration (160 mg/5 mL).**

Dosing for Infants and Children

From Your Healthcare Professional

Concentrated
Motrin
Infants' Drops

Children's Motrin

DOSE: Every 6-8 hours as needed. DO NOT GIVE MORE THAN 4 DOSES IN 24 HOURS.

If possible, use weight to dose; otherwise use age.

mL = milliliter
tsp = teaspoon



Infants' MOTRIN® Concentrated Drops

Active Ingredient:
Ibuprofen 50 mg (NSAID)[†]
(in each 1.25 mL)

[†]Nonsteroidal anti-inflammatory drug
Use only as directed.

Available in:
 Dye-Free Berry



Children's MOTRIN® Oral Suspension

Active Ingredient:
Ibuprofen 100 mg (NSAID)[†]
(in each 5 mL or 1 tsp)

[†]Nonsteroidal anti-inflammatory drug
Use only as directed.

Available in:
 Original Berry
 Dye-Free Berry

WEIGHT	AGE		
6-11 lbs	0-5 mos	Do not use	—
12-17 lbs	6-11 mos	1.25 mL	—
18-23 lbs	12-23 mos	1.875 mL	—
24-35 lbs	2-3 yrs	—	5 mL (1 tsp)
36-47 lbs	4-5 yrs	—	7.5 mL (1½ tsp)
48-59 lbs	6-8 yrs	—	10 mL (2 tsp)
60-71 lbs	9-10 yrs	—	12.5 mL (2½ tsp)
72-95 lbs	11 yrs	—	15 mL (3 tsp)

IMPORTANT INSTRUCTIONS for Proper Use

Today's Date: _____

This dosing recommendation from your doctor will expire in 14 DAYS.

- Read and follow the label on all MOTRIN® products.
- Take every 6-8 hours as needed. Do NOT exceed more than 4 doses in 24 hours.
- Do NOT administer longer than 10 days, unless directed by a doctor.
- Ask a doctor or pharmacist before use if the child is taking any other drug containing an NSAID[†] (prescription or non-prescription).
 - MOTRIN® contains ibuprofen
- Use only the dosing device that comes with a specific product.

[†]Nonsteroidal anti-inflammatory drug

Pediatric Specific Medication Notes

- Antibiotic
selection
- Analgesics



Antibiotics

- Do **not** prescribe antibiotics if you are uncertain if there is an infection
- Antibiotics generally do not penetrate to teeth
- If the dental infection is contained within the dental tissues antibiotics are usually not indicated



Antibiotics

- First line antibiotics should be broad spectrum and widely accepted
- Amoxicillin or Penicillin VK are our first choices



Antibiotics

- Be *extremely* judicious with use of more intensive antibiotics – Amoxicillin-Clavulanate (Augmentin) should only be prescribed when a first choice is unsuccessful
- Clindamycin should be a last resort



Analgesics

- Acetaminophen or Ibuprofen are the primary agents for children
- Aspirin should be avoided due to the possibility of Reyes syndrome



Reyes Syndrome

Condition resulting in: nausea and vomiting, tachypnea, sluggishness, irritability, anxiety, seizures, and coma

Often occurs in children and teens

Cause is unknown – but linked to taking salicylate (aspirin) during or immediately after a viral infection

Analgesics

- Use extreme caution with opioids of *any* type in children
- There is a historic preference for Tylenol #3 (acetaminophen/codeine) in pediatric use
- Variable metabolism of codeine makes this a poor choice
- Do NOT prescribe codeine for children under 12

Writing a Pediatric Script

1

Obtain a
current patient
weight

2

Select an
appropriate
medication

3

Identify the
proper dose
and duration

4

Identify dosage
forms

5

Calculate the
patient dose

6

Write the
script

Demo

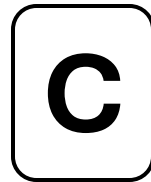
- Patient is a 6 year old male with asthma
- She has an intraoral swelling associated with tooth #S and Dr. Marion and Dr. Yun are on vacation for 2 weeks so we can't take it out until they get back

Write our Script

Rx:



Sig:



Disp:



Obtain a current weight

Our patient
weighs 53lbs or
24 kg





53lbs /
24kg

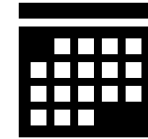
Select an Appropriate Medication



Infection → Bacterial → Antibiotic



First line choice → Amoxicillin



24kg

Amoxicillin

Identify the proper dose and duration

- Don't be afraid to look this up
- Amoxicillin Dose for Children older than 3 and below 40 kg
 - 20-40mg/kg/day *in divided doses* every 8 hours
 - 25-45mg/kg/day *in divided doses* every 12 hours
- Duration 10-14 days



24kg

Amoxicillin

**30mg/kg/day
in 3 doses**

10 days

Next Step

- How is this patient going to take this medication?
- Most kids do not do pills well
- Many adults don't either



Medication Formulary - Amoxicillin

- Capsule
 - 250mg
 - 500mg
- Tablet
 - 500mg
 - 875mg
- Oral suspension
 - 125mg/5mL
 - 200mg/5mL
 - 250mg/5mL
 - 400mg/5mL





24kg

**Amoxicillin
250mg/5mL**

**30mg/kg/day
in 3 doses**

10 days



÷ 5

250mg/5mL

50mg/mL

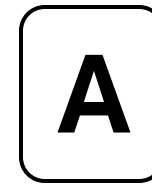


24kg

50mg/mL

**30mg/kg/day
in 3 doses**

10 days



24kg

**30mg/kg/day
in 3 doses**

**720mg/day in 3
divided doses**

Convert the Dose to Liquid Units



24kg

**Amoxicillin
250mg/50mL**

**30mg/kg/day
in 3 doses**

10 days

A



B

**720mg/day in 3
divided doses**

50mg/mL

**14.4mL/day in 3
divided doses**

Convert the Dose to Liquid Units



24kg

**Amoxicillin
250mg/50mL**

**30mg/kg/day
in 3 doses**

10 days

B



3

=

C

14.4mL/day

**in 3 divided
doses**

4.8mL dose

Determine total medication needed



24kg

**Amoxicillin
250mg/50mL**

**30mg/kg/day
in 3 doses**

10 days

B



D

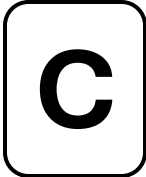
**14.4mL/day in 3
divided doses**

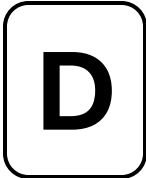
7 days

**100.8mL
total**

Write our Script

Rx: 

Sig:   

Disp: 

Write our Script

Rx: Amoxicillin 250mg/5mL solution

Sig: Take 4.8mL by mouth three times a day for 10 days

Disp: 144 mL

Most are simpler

- We will also give this patient an acetaminophen script for pain
- Acetaminophen dosing 10-15 mg/kg/dose every 4-6 hours as needed
- Acetaminophen liquid is 160mg/5mL
- We will give this for 10 days to match our antibiotic



24kg

**Acetaminophen
160mg/5mL
liquid**

**10-15mg/kg/
dose**

**Q4-6H prn
pain**



24kg

**Acetaminophen
32mg/1mL
liquid**

**10 mg/kg/
dose**

**Q4-6H prn
pain**

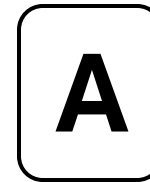


24kg

**Acetaminophen
32mg/1mL
liquid**

**10 mg/kg/
dose**

**Q4-6H prn
pain**



24kg

**10mg/kg/
dose**

240mg/dose

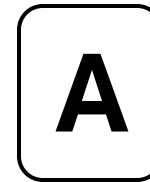


24kg

Acetaminophen
32mg/1mL
liquid

10 mg/kg/
dose

Q4-6H prn
pain



24kg

10mg/kg/
dose

240mg/dose



24kg

Acetaminophen
32mg/1mL
liquid

10 mg/kg/
dose

Q4-6H prn
pain

A



C

240mg/dose

32mg/mL

7.5mL/ dose



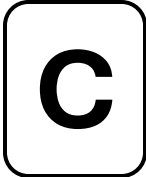
24kg	Acetaminophen 32mg/1mL liquid	10 mg/kg/ dose	Q4-6H prn pain for 10 days
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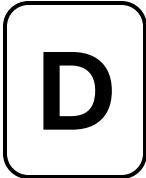
$$\boxed{C} \times \boxed{6} \times \boxed{10} = \boxed{D}$$

7.5mL/ dose	Doses/ day	Days	450 mL
----------------	---------------	------	--------

Write our Script

Rx: 

Sig:   

Disp: 

Write our Script

Rx: Acetaminophen 160/5mL
solution

Sig: Take 7.5mL by mouth every 4-6
hours as needed for pain

Disp: 450mL

Wrap Up

- Local anesthetic properties help guide their specific use in children
- Although principles are the same – children require modifications to technique for effective anesthetic delivery
- Dosage of medications (including LA) are almost always weight based – and it is vital to follow this to avoid overdose



Questions